

The Complex Integral

Complex Analysis

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Contents

Morales C. Ronny

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Gauges

The Gauge Integral

Properties of $\int_a^b$ The Fundamental
Theorem (Calculus)The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$ The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral
FormulaThe Fundamental
Theorem of Algebra

Taylor's Theorem

Definition

We say that an interval in $\mathbb{R}$ is **degenerate** [respectively, **nondegenerate**] if it contains at most one point [respectively, if it contains at least two points]. Two intervals are disjoint if their intersection is empty, similarly, we say that two intervals are **nonoverlapping** if their intersection is degenerate.

Definition

If $I \doteq [a, b]$ with $a \leq b$, we define the **length** of I to be

$$\ell(I) \doteq b - a.$$

Similarly, the length of any interval having one of the three forms:

$$(a, b), \quad [a, b), \quad (a, b]$$

is also defined to be equal to $b - a$. In particular, $\ell(\emptyset) = 0$.

Definition

If $I \doteq [a, b]$ is a closed, bounded interval in $\mathbb{R}$, then a **partition** of I is a finite, ordered set $\mathcal{P} \doteq (x_0, x_1, \dots, x_{n-1}, x_n)$ of points in I such that

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b.$$

The points of $\mathcal{P}$ are used to divide $I = [a, b]$ into non-overlapping subintervals

$$I_1 \doteq [x_0, x_1], \quad I_2 \doteq [x_1, x_2], \dots, \quad I_n \doteq [x_{n-1}, x_n].$$

Often we will write $\mathcal{P} = \{[x_{i-1}, x_i]\}_{i=1}^n$. If a point has been selected from each subinterval $I_i = [x_{i-1}, x_i]$, for $i = 1, \dots, n$, then the points are called **tags** of the subintervals I_i . A set of ordered pairs

$$\mathcal{P}^* \doteq \{(I_i, x_i^*)\}_{i=1}^n$$

of subintervals and corresponding tags is called a **starred partition** of I .

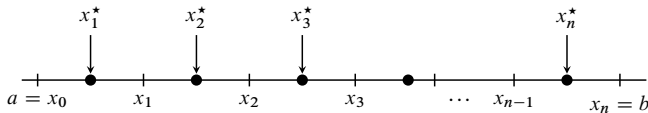


Figure 1: A starred partition of $[a, b]$

Contents

Morales C. Ronny

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

A **gauge** on I is a strictly positive function defined on I . If δ is a gauge on I , then we say that a starred partition $\mathcal{P}^*$ is **δ -fine** if

$$x_i^* \in I_i \subseteq [x_i^* - \delta(x_i^*), x_i^* + \delta(x_i^*)] \quad \text{for } i = 1, \dots, n.$$

In this case we write $\mathcal{P}^* \ll \delta$. For the sake of economy of notation we will write $\overline{V}_\delta(x_i^*) \equiv [x_i^* - \delta(x_i^*), x_i^* + \delta(x_i^*)]$.

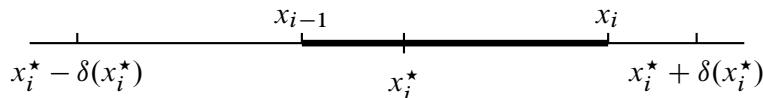


Figure 2: Inclusion in the previous definition

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Lemma 1.1

If a starred partition $\mathcal{P}^$ of $I \doteq [a, b]$ is δ -fine and $x \in I$, then there exists a tag x_i^* in $\mathcal{P}^*$ such that $x \in \overline{V}_\delta(x_i^*)$.*

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Remark

If δ and γ are gauges on $I \doteq [a, b]$ and if $0 < \delta(x) \leq \gamma(x)$ for all $x \in I$, then $\mathcal{P}^* \ll \delta$ implies $\mathcal{P}^* \ll \gamma$. This follows from

$$x_i^* - \gamma(x_i^*) \leq x_i^* - \delta(x_i^*) \quad \text{and} \quad x_i^* + \delta(x_i^*) \leq x_i^* + \gamma(x_i^*)$$

which imply that

$$x_i^* \in \overline{V}_\delta(x_i^*) \subseteq \overline{V}_\gamma(x_i^*) \quad \text{for } i = 1, \dots, n.$$

Moreover, if δ_1, δ_2 are gauges on I , then $\delta \doteq \inf\{\delta_1, \delta_2\}$ is also a gauge on I and any δ_1 -fine (or δ_2 -fine) partition is also δ -fine by the previous result.

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Example 1.2

a) Let $I \doteq [0, 1]$ and define

$$\delta = \begin{cases} \frac{1}{10} & x = 0, \\ \frac{1}{2}x & 0 < x \leq 1. \end{cases}$$

It is readily seen that δ is a gauge on I . If $0 < x^* \leq 1$, then $V_\delta(x^*) = [\frac{1}{2}x^*, \frac{3}{2}x^*]$, which does not contain the point 0. Thus, if $\mathcal{P}^*$ is a δ -fine partition of I , then the only subinterval in $\mathcal{P}^*$ that contains 0 must have the point 0 as its tag.

b) Let γ be defined on $I \doteq [0, 1]$ by

$$\gamma(x) \doteq \begin{cases} \frac{1}{10} & x \in \{0, 1\}, \\ \frac{1}{2}x & 0 < x \leq 1/2, \\ \frac{1}{2}(1-x) & \frac{1}{2} < x < 1. \end{cases}$$

Then γ is a gauge on I , and the subintervals in any γ -fine partition that contain the points 0 or 1 must have these points as tags.

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Theorem 1.3 (Cousin's Theorem)

If δ is a gauge defined on the nondegenerate interval $[a, b]$, then there exists a δ -fine partition of $[a, b]$.

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

If $\mathcal{P}^\star = \{(I_j, t_j)\}_{j=1}^n$ is a starred partition of $[a, b]$, we define the **Riemann sum** of a function $f : [a, b] \rightarrow \mathbb{C}$ to be the number

$$S(f; \mathcal{P}^\star) \doteq \sum_{j=1}^n f(t_j)(x_j - x_{j-1}).$$

Definition

A function $f : [a, b] \rightarrow \mathbb{C}$ is said to be **Henstock-Kurzweil integrable**[†] on $[a, b]$ if there exists a number $L \in \mathbb{C}$ such that for every $\varepsilon > 0$ there exists a gauge δ_ε on $[a, b]$ such that if $\mathcal{P}^\star$ is any δ_ε -fine partition of $[a, b]$, then

$$|S(f; \mathcal{P}^\star) - L| < \varepsilon.$$

The family of all Henstock-Kurzweil integrable functions will usually be denoted by $\mathcal{HK}[a, b]$.

[†]Due to the similarities between this and the standard Riemann integral, this is sometimes called the **generalised Riemann integral** and said that f is **generalised Riemann integrable**. The family of the latter is denoted by $\mathcal{R}^*[a, b]$.

Theorem 1.4 (Uniqueness Theorem)

If $f \in \mathcal{HK}[a, b]$, then the value of L is uniquely determined.

Definition

If $f \in \mathcal{HK}[a, b]$, then the unique value L in the previous definition is called the **Henstock-Kurzweil integral** of f over $[a, b]$. Instead of L , we will usually write

$$L = \int_a^b f \quad \text{or} \quad \int_a^b dx \, f(x).$$

It should be understood that any letter other than x can be used in the latter expression, so long as it does not cause any ambiguity.

Example 1.5

The **Dirichlet function** $f : [0, 1] \rightarrow \mathbb{R}$, defined as

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q}, \\ 0 & x \notin \mathbb{Q}, \end{cases}$$

belongs to $\mathcal{HK}[0, 1]$ and has integral 0.

Theorem 1.6

If $g \in \mathcal{HK}[a, b]$ and if $f(x) = g(x)$ except for a finite number of points in $[a, b]$, then f is in $\mathcal{HK}[a, b]$ and

$$\int_a^b f = \int_a^b g.$$

Theorem 1.7 (Properties of the Henstock-Kurzweil Integral)

Suppose that $f, g \in \mathcal{HK}[a, b]$. Then:

i) If $k \in \mathbb{C}$, the function $kf + g$ is in $\mathcal{HK}[a, b]$ and

$$\int_a^b (kf + g) = k \int_a^b f + \int_a^b g.$$

ii) If $f^*(x) \doteq (f(x))^*$, then the function f^* is in $\mathcal{HK}[a, b]$ and

$$\int_a^b f^* = \left(\int_a^b f \right)^*.$$

iii) If $\Im(f) = \Im(g) = 0$ and $f \leq g$ on $[a, b]$, then

$$\int_a^b f \leq \int_a^b g.$$

Theorem 1.8 (Cauchy Criterion)

A function $f : [a, b] \rightarrow \mathbb{C}$ belongs to $\mathcal{HK}[a, b]$ if and only if for every $\varepsilon > 0$ there exists a gauge η_ε on $[a, b]$ such that if $\mathcal{P}^\star$ and $\mathcal{Q}^\star$ are starred partitions of $[a, b]$ that are η_ε -fine, then

$$|S(f; \mathcal{P}^\star) - S(f; \mathcal{Q}^\star)| < \varepsilon.$$

Theorem 1.9 (Squeeze Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$. Then $f \in \mathcal{HK}[a, b]$ if and only if for every $\varepsilon > 0$ there exist functions α_ε and ω_ε in $\mathcal{HK}[a, b]$ with

$$\alpha_\varepsilon \leq f \leq \omega_\varepsilon \quad \text{on } [a, b],$$

and such that

$$\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon.$$

Theorem 1.10 (Additivity Theorem)

Let $f : [a, b] \rightarrow \mathbb{C}$ and let $c \in (a, b)$. Then $f \in \mathcal{HK}[a, b]$ if and only if its restrictions to $[a, c]$ and $[c, b]$ are both Henstock-Kurzweil integrable. In this case

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Corollary 1.11

If $f \in \mathcal{HK}[a, b]$ and if $\mathcal{P} = \{c_i\}_{i=0}^n$ is a partition of $[a, b]$, then the restrictions of f to each of the subintervals $[c_{i-1}, c_i]$ are in $\mathcal{HK}[a, b]$ and

$$\int_a^b f = \sum_{i=1}^n \int_{c_{i-1}}^{c_i} f.$$

Definition

If $f \in \mathcal{HK}[a, b]$ and $\alpha, \beta \in [a, b]$ with $\alpha < \beta$, we define

$$\int_{\beta}^{\alpha} f \doteq - \int_{\alpha}^{\beta} f \quad \text{and} \quad \int_{\alpha}^{\alpha} f \doteq 0.$$

Theorem 1.12

If $f \in \mathcal{HK}[a, b]$ and if α, β, γ are any numbers in $[a, b]$, then

$$\int_{\alpha}^{\beta} f = \int_{\alpha}^{\gamma} f + \int_{\gamma}^{\beta} f,$$

in the sense that the existence of any two of these integrals implies the existence of the third integral and the previous equality.

Theorem 1.13 (Fundamental Theorem of Calculus (First Form))

Suppose there exists a countable set E in $[a, b]$, and functions $f, F : [a, b] \rightarrow \mathbb{C}$ such that:

- i) F is continuous on $[a, b]$.
- ii) $F'(x) = f(x)$ for all $x \in [a, b] \setminus E$.

Then f belongs to $\mathcal{HK}[a, b]$ and

$$\int_a^b f = F(b) - F(a).$$

Definition

If $f \in \mathcal{HK}[a, b]$, then the function defined by

$$F(x) \doteq \int_a^x f \quad \text{for } x \in [a, b],$$

is called the **indefinite integral** of f with **basepoint** a .

Definition

A set $Z \subset \mathbb{R}$ is said to be a **null set** if for every $\varepsilon > 0$ there exists an open countable cover $\{I_n\}_{n \in \mathbb{N}}$ of Z such that

$$\sum_{n \in \mathbb{N}} \ell(I_n) \leq \varepsilon.$$

If $Q(x)$ is a statement about the point $x \in I$, we say that $Q(x)$ holds **almost everywhere** on I (or for **almost every** $x \in I$), if there exists a null set $Z \subset I$ such that $Q(x)$ holds for all $x \in I \setminus Z$. In this case we write

$$Q(x) \quad \text{for a.e. } x \in I.$$

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Example 1.14

Show that the set $\mathbb{Q}$ of rational numbers is a null set.

Theorem 1.15 (Fundamental Theorem of Calculus (Second Form))

Let f belong to $\mathcal{HK}[a, b]$ and let F be the indefinite integral of f . Then we have:

- i) F is continuous on $[a, b]$.*
- ii) F is differentiable at x and $F'(x) = f(x)$ for a.e. $x \in I$.*
- iii) If f is continuous at $c \in [a, b]$, then $F'(c) = f(c)$.*

Theorem 1.16 (Substitution Theorem)

- i) Let $I \doteq [a, b]$ and $J \doteq [\alpha, \beta]$, and let $F : I \rightarrow \mathbb{C}$ and $\varphi : J \rightarrow \mathbb{C}$ be continuous functions with $\varphi(J) \subseteq I$.
- ii) Suppose that there exist sets $E_f \subset I$ and $E_\varphi \subset J$ such that $f(x) = F'(x)$ for $x \in I \setminus E_f$, that $\varphi'(t)$ exists for $t \in J \setminus E_\varphi$, and that $E \doteq \varphi^{-1}(E_f) \cup E_\varphi$ is countable.
- iii) Set $f(x) \doteq 0$ for $x \in E_f$ and $\varphi'(t) \doteq 0$ for $t \in E_\varphi$.

We conclude that $f \in \mathcal{HK}(\varphi(J))$, that $(f \circ \varphi)\varphi' \in \mathcal{HK}(J)$ and that

$$\int_\alpha^\beta (f \circ \varphi)\varphi' = F \circ \varphi \Big|_\alpha^\beta = \int_{\varphi(\alpha)}^{\varphi(\beta)} f.$$

Example 1.17*Show that*

$$\int_0^1 dt \frac{1}{\sqrt{t-t^2}} = \pi.$$

Theorem 1.18 (Integration by Parts Theorem)

Let F and G be differentiable on $[a, b]$. Then $F'G$ belongs to $\mathcal{HK}[a, b]$ if and only if FG' belongs to $\mathcal{HK}[a, b]$. In this case we have

$$\int_a^b F'G = FG \Big|_a^b - \int_a^b FG'.$$

Theorem 1.19 (Multiplication Theorem)

If $f \in \mathcal{HK}[a, b]$ and if g is a (real) monotone function on $[a, b]$, then the product fg belongs to $\mathcal{HK}[a, b]$.

Theorem 1.20 (Taylor's Theorem)

Suppose that all the derivatives of f up to order $n + 1$ exist on $[a, b]$. Then we have

$$f(b) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (b-a)^k + R_n,$$

where the remainder is given by

$$R_n = \frac{1}{n!} \int_a^b dt f^{(n+1)}(t) \cdot (b-t)^n.$$

Definition

A function $f \in \mathcal{HK}[a, b]$ such that $|f| \in \mathcal{HK}[a, b]$ is said to be **Lebesgue integrable** on $[a, b]$. The collection of all Lebesgue integrable functions on $[a, b]$ is denoted by $\mathcal{L}[a, b]$.

Definition

A **subpartition** of $[a, \infty)$ is a finite collection of non-overlapping intervals of finite length whose union is properly contained in $[a, \infty)$. We define a **gauge** on $[a, \infty)$ to be an ordered pair consisting of a strictly positive function δ defined on $[a, \infty)$ and a number $d^* > 0$. When we say that a starred subpartition $\mathcal{P}^*$ is **δ -fine**, we mean that

$$[a, \infty) = \bigcup_{i \in \mathbb{N}} [x_{i-1}, x_i] \cup [x_n, \infty),$$

that

$$[x_{i-1}, x_i] \subseteq [x_i^* - \delta(x_i^*), x_i^* + \delta(x_i^*)] \quad \text{for } i = 1, \dots, n,$$

and that

$$[x_n, \infty) \subseteq [1/d^*, \infty)$$

or, equivalently, that

$$\frac{1}{d^*} \leq x_n.$$

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

- i) A function $f : [a, \infty) \rightarrow \mathbb{C}$ is said to be **Henstock-Kurzweil integrable** if there exists $A \in \mathbb{C}$ such that for every $\varepsilon > 0$ there exists a gauge δ_ε on $[a, \infty]$ such that if $\mathcal{P}^*$ is any δ_ε -fine starred subpartition of $[a, \infty)$, then

$$|S(f; \mathcal{P}^*) - A| \leq \varepsilon.$$

In this case we write $f \in \mathcal{HK}[a, \infty)$ and

$$\int_a^\infty f = A.$$

- ii) A function is said to be **Lebesgue integrable** if both f and $|f|$ belong to $\mathcal{HK}[a, \infty)$. In this case we write $f \in \mathcal{L}[a, \infty)$.

Definition

Let $b \in \mathbb{R}$ and $g : (-\infty, b] \rightarrow \mathbb{C}$. A **gauge** on $[-\infty, b]$ is an ordered pair consisting of a number $d_* > 0$ and a strictly positive function δ on $(-\infty, b]$. When we say that a starred subpartition $\mathcal{P}^*$ of $(-\infty, b]$ is **δ -fine**, we mean that

$$(-\infty, b] = (-\infty, x_0] \cup \bigcup_{i \in \mathbb{N}} [x_{i-1}, x_i],$$

that

$$[x_{i-1}, x_i] \subseteq [x_i^* - \delta(x_i^*), x_i^* + \delta(x_i^*)] \quad \text{for } i = 1, \dots, n,$$

and that

$$(-\infty, x_0] \subseteq (-\infty, -1/d_*]$$

or, equivalently, that

$$x_0 \leq -1/d_*.$$

Definition

- i) A function $f : (-\infty, b] \rightarrow \mathbb{C}$ is said to be **Henstock-Kurzweil integrable** if there exists $B \in \mathbb{C}$ such that for every $\varepsilon > 0$ there exists a gauge δ_ε on $(-\infty, b]$ such that if $\mathcal{P}^*$ is any δ_ε -fine starred subpartition of $(-\infty, b]$, then

$$|S(f; \mathcal{P}^*) - B| \leq \varepsilon.$$

In this case we write $f \in \mathcal{HK}(-\infty, b]$ and

$$\int_{-\infty}^b f = B.$$

- ii) A function is said to be **Lebesgue integrable** if both f and $|f|$ belong to $\mathcal{HK}(-\infty, b]$. In this case we write $f \in \mathcal{L}(-\infty, b]$.

Definition

1. A **curve** is a map $\gamma : I \subseteq \mathbb{R} \rightarrow \mathbb{C}$ from the real interval $I \doteq [a, b]$ into the complex plane; $\gamma(a)$ is called the **initial point** of the curve and $\gamma(b)$ its **final point**, if the former and the latter coincide, then the curve is said to be **closed**. Moreover, γ is said to **join** its initial and final points. The range $\gamma^\dagger \equiv \gamma(I) \subset \mathbb{C}$ is called the **track** of γ , and if $\gamma^\dagger \subset \Omega \subseteq \mathbb{C}$, then we say that γ is a curve in Ω .

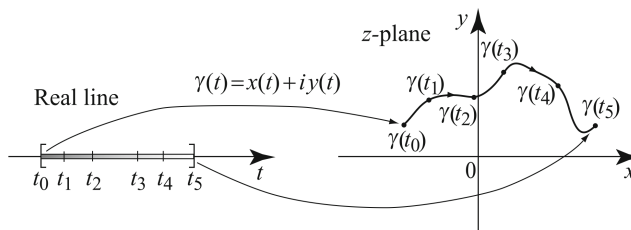


Figure 3: Mapping from a real interval I into the complex plane

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

2. A **simple curve** is a curve that does not self-intersect, i.e., γ is injective, so that $\gamma(t_1) \neq \gamma(t_2)$ when $t_1 \neq t_2$. When the curve is closed and simple except for the fact that $\gamma(a) = \gamma(b)$, then it is called a **Jordan curve**.
3. A **path** is a finite collection $\{\gamma_i\}_{i=1}^n$ of simple curves such that the initial point of γ_{k+1} coincides with the final point of γ_k . If each γ_i is a simple curve in Ω , then we say that the path is in Ω .
4. A **smooth curve** is a continuously differentiable curve for which $\gamma'(t) \neq 0$ for $t \in I$.
5. A **contour** is a path whose curves are smooth. When the initial point of γ_1 coincides with the final point of γ_n , the contour is said to be a **closed contour**. If the path is in Ω , we say that the contour is in Ω .

Remark

By derivative at the endpoint a [respectively, b] we mean the appropriate sided derivative, namely

$$\gamma'(a) = \lim_{z \rightarrow a^+} \frac{\gamma(z) - \gamma(a)}{z - a}, \quad \left[\text{respectively, } \lim_{z \rightarrow b^-} \frac{\gamma(z) - \gamma(b)}{z - b} \right].$$

Theorem (Jordan Curve Theorem)

Let γ be a Jordan curve. Then $\gamma^\dagger$ divides the complex plane into an “interior” region (the unique path-connected component of $\mathbb{C} \setminus \gamma^\dagger$ that is bounded) denoted by $\text{int}(\gamma)$, bounded by the track of γ , and an “exterior” region (the unique path-connected component of $\mathbb{C} \setminus \gamma^\dagger$ that is unbounded), denoted by $\text{ext}(\gamma)$, i.e.

$$\mathbb{C} \setminus \gamma^\dagger = \text{int}(\gamma) \coprod \text{ext}(\gamma).$$

Definition

*Since the interior and the exterior regions relative to a Jordan curve are identifiable, we may define the orientations of such a curve. The **positive orientation** of a simple closed curve is the one that places the interior region to our left when traced. The **negative orientation** is the one that, when traced, places the interior region to our right.*

Remark

Let $\mathcal{P} = \{[t_{k-1}, t_k]\}_{k=1}^n$ be a partition of $I \doteq [a, b]$ and let γ be a smooth curve from I into $\mathbb{C}$. We can approximate the length of γ via sums of the form

$$\sum_{k=1}^n |\gamma(t_k) - \gamma(t_{k-1})|.$$

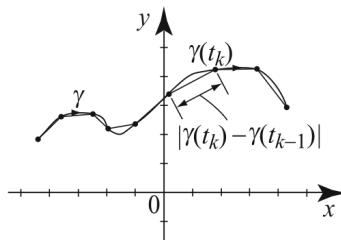


Figure 4: Sum approximation for the length of a curve γ

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

Let $\{\gamma_i\} = \gamma : I \subseteq \mathbb{R} \rightarrow \mathbb{C}$ be a contour and let $\gamma_i : I_i \subset I \rightarrow \mathbb{C}$ its simple smooth curves. We define the **length** of γ to be the number

$$L(\gamma) \doteq \sum_{j=1}^n \int_{I_j} |\gamma_j'|.$$

Example 1.21

Find the arc length of a **cycloid** with radius $a > 0$ defined by

$$\gamma(t) = a(t - \sin t) + ia(1 - \cos t), \quad t \in [0, 2\pi].$$

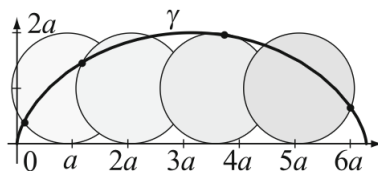


Figure 5: Arc length of a cycloid

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a contour in Ω and let $f : \Omega \rightarrow \mathbb{C}$ be continuous on the track of γ . Then the **contour integral of f along γ** is defined to be the number

$$I \doteq \int_a^b (f \circ \gamma) \gamma'.$$

Instead of I , we will usually write

$$I \equiv \int_{\gamma} f \quad \text{or} \quad I \equiv \int_{\gamma} dz f(z).$$

If Γ is a closed contour, the notation

$$\oint_{\Gamma} f$$

is usually used to express the fact that the integration is along a closed contour.

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour IntegrationProperties of $\int_{\gamma}$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

According to the previous notation, we define the **contour integral of f along γ with respect to z^*** to be the number

$$\int_\gamma dz^* f \doteq \left(\int_\gamma dz f^* \right)^*.$$

Example 1.22

Let $w_1, w_2 \in \mathbb{C}$ and let $\gamma : [0, 1] \rightarrow \mathbb{C}$ be the line segment joining w_1 to w_2 . Then compute

$$\int_\gamma dz \, z \quad \text{and} \quad \int_\gamma dz^* \, z^*.$$

Definition

Let $R > 0$ and let $z_0 \in \mathbb{C}$. The particular closed contour $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$ defined by $\gamma(t) = z_0 + Re^{it}$ is called the **positively oriented circle with centre z_0 and radius R** . The notation

$$\oint_{B_R(z_0)} dz f(z)$$

is sometimes used to denote the integral of f along such contour.

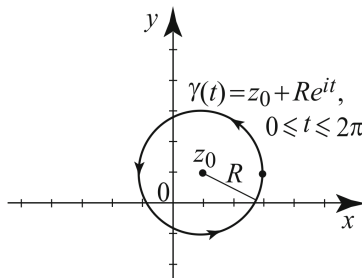


Figure 6: Positively oriented circle with centre z_0 and radius R

Contents

Morales C. Ronny

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Example 1.23

Let $r > 0$ and hence compute the integral

$$\oint_{B_R(z_0)} dz (z - z_0)^n$$

for the different values of $n = -2, -1, 0, 1, 2$.

Theorem 1.24 (Darboux Inequality)

Suppose f is continuous and bounded on $\gamma^\dagger$, i.e., there exists a positive number M such that $|f(\gamma^\dagger)| \leq M$. Then

$$\left| \int_\gamma f \right| \leq ML(\gamma).$$

More generally

$$\left| \int_\gamma f \right| \leq \int_a^b |f \circ \gamma| |\gamma'|.$$

Gauges

The Gauge Integral

Properties of $\int_a^b$ The Fundamental
Theorem (Calculus)The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$ The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral
FormulaThe Fundamental
Theorem of Algebra

Taylor's Theorem

Definition

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a curve. The curve **opposite** to γ is defined as

$$\gamma^-(t) \doteq \gamma(a + b - t)$$

for $t \in [a, b]$.

Theorem 1.25

Let γ be a contour and let $f, g : \gamma^\dagger \rightarrow \mathbb{C}$ be continuous. Then for all $\lambda \in \mathbb{C}$,

$$\int_{\gamma} (\lambda f + g) = \lambda \int_{\gamma} f + \int_{\gamma} g,$$

and if γ^- is the contour opposite to γ , then

$$\int_{\gamma} f = - \int_{\gamma^-} f.$$

In particular, if Γ is a closed contour, then

$$\oint_{\Gamma} f = - \oint_{\Gamma} f.$$

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_{\gamma}$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

Let $\gamma : [a, b] \rightarrow \mathbb{C}$, $\sigma : [\alpha, \beta] \rightarrow \mathbb{C}$ be curves. We say that σ is a **reparametrisation** of γ if there is a continuously differentiable bijection ϕ of $[\alpha, \beta]$ onto $[a, b]$ with an everywhere strictly positive derivative ϕ' , such that $\sigma = \gamma \circ \phi$. Similarly, if $\Gamma : I \subset \mathbb{R} \rightarrow \mathbb{C}$, $\Gamma = \{\Gamma_i\}_{i=1}^n$ and $\Sigma : J \subset \mathbb{R} \rightarrow \mathbb{C}$, $\Sigma = \{\Sigma_i\}_{i=1}^n$ are contours, then we say that Σ is a **reparametrisation** of Γ if Σ_i is a reparametrisation of Γ_i for each $i = 1, \dots, n$.

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental
Theorem (Calculus)

The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral
Formula

The Fundamental
Theorem of Algebra

Taylor's Theorem

Proposition 1.26

If Σ is a reparametrisation of Γ , then Σ and Γ have the same track.

Theorem 1.27

Let Σ be a reparametrisation of a contour Γ and let $f : \Gamma^\dagger \rightarrow \mathbb{C}$ be continuous. Then

$$\int_{\Sigma} f = \int_{\Gamma} f.$$

Definition

Let G be an open subset of $\mathbb{C}$ and let $f : G \rightarrow \mathbb{C}$. A function $F \in \mathcal{H}(G)$ is called a **primitive** (or **antiderivative**) for f on G if $F' = f$.

Theorem 1.28 (The Fundamental Theorem of Contour Integration)

Suppose that f is continuous on a domain Ω . Then the following statements are equivalent:

- a) f has a primitive on Ω ;
- b) the integral of f along any closed contour in Ω vanishes;
- c) if γ and μ are two contours in Ω joining z_1 to z_2 , then the **Newton-Leibniz formula** holds:

$$\int_\gamma f = \int_\mu f = F(z_2) - F(z_1),$$

where F is the primitive from (a).

Notation

Due to the fact that if any of the previous statements is true, then the integral along a contour joining z_1 to z_2 is independent of the contour chosen, we will sometimes write

$$\int_{z_1}^{z_2} f$$

to denote the value of such integral.

Theorem 1.29 (Green's Theorem)

Let γ be a Jordan curve with positive orientation and D be the region interior to γ . Suppose that $P(x, y)$ and $Q(x, y)$ are at least class $C^1(\overline{D})$. Then

$$\int_\gamma dx P + dy Q = \iint_D dx dy \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right).$$

Theorem 1.30 (Cauchy's Theorem)

Suppose that f is in $\mathcal{H}(\Omega)$ with Ω a region that contains a Jordan curve Γ in its interior. If f' is continuous on Ω , then we have

$$\oint_{\Gamma} f = 0.$$

Lemma (Goursat's Lemma) 1.31

Let f be a continuous function on an open set U in $\mathbb{C}$ and assume that $f'(z_0)$ exists for some z_0 in U . Then for given $\varepsilon > 0$ there exists $\delta > 0$ such that for every closed triangle Δ that contains z_0 with $\Delta \subset B_\delta(z_0) \cap U$ we have

$$\left| \int_{\partial\Delta} f \right| < \varepsilon (\text{diam}(\Delta)).$$

Theorem 1.32 (Cauchy-Goursat Theorem)

Suppose that Δ is a closed triangle contained in a region Ω and let $f \in \mathcal{H}(\Omega)$. Let $\partial^+ \Delta$ be the boundary of Δ with positive orientation. Then we have

$$\int_{\partial^+ \Delta} f = 0.$$

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Remark

We now wish to extend Cauchy's theorem to more general regions. The approach we take is based on the geometric concepts of deformation of paths (the concept of homotopic curves) that is easy to visualise, as illustrated by some examples. A precise definition is motivated by the following three cases.

Case I

Suppose that α and β are points in a region Ω , and γ_0 and γ_1 are contours in Ω joining α to β . It may be possible to move γ_0 continuously so that it coincides with γ_1 , keeping the endpoints fixed at α and β and without leaving Ω .

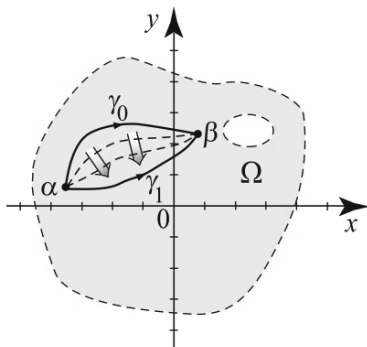


Figure 7: Continuous deformation of γ_0 into γ_1

Contents

Morales C. Ronny

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Case II

If γ_0 and γ_1 are two closed contours in a region Ω , it may be possible to continuously move γ_0 without leaving Ω , in such way that it coincides with γ_1 in position and direction.

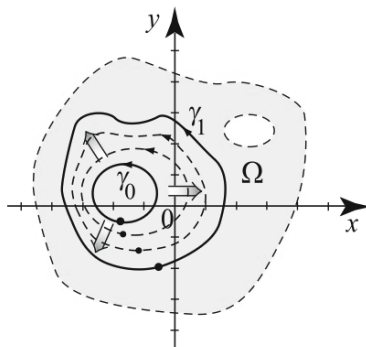


Figure 8: Continuous deformation of γ_0 into γ_1

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Case III

If γ_0 is a closed contour in a region Ω , it may be the case that we can continuously shrink γ_0 to a point $z_0 \in \Omega$ without leaving Ω . This situation is a special case of the preceding one, when $\gamma_1(t) = z_0$ for all t , i.e., it is degenerated into a point.

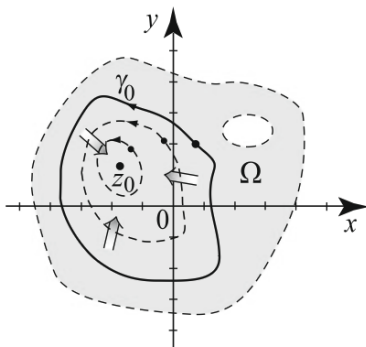


Figure 9: Continuous deformation of γ_0 into a point z_0

Contents

Morales C. Ronny

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

Let γ_0, γ_1 be contours in a region Ω , both defined on $[0, 1]$. Assume that $\gamma_0(0) = \gamma_1(0) = \alpha \in \Omega$ and $\gamma_0(1) = \gamma_1(1) = \beta \in \Omega$. We say that γ_0 is **homotopic** to γ_1 (**relative to Ω**), and write $\gamma_0 \approx \gamma_1$, if there exists a continuous map H from the unit square $Q = [0, 1] \times [0, 1]$ to Ω with the following properties:

$$H(t, 0) = \gamma_0(t), \quad 0 \leq t \leq 1;$$

$$H(t, 1) = \gamma_1(t), \quad 0 \leq t \leq 1;$$

$$H(0, s) = \alpha, \quad 0 \leq s \leq 1;$$

$$H(1, s) = \beta, \quad 0 \leq s \leq 1.$$

Definition

The continuous mapping H is called a **homotopy** (or a **continuous deformation**).

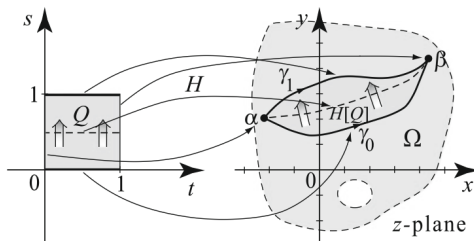


Figure 10: Continuous deformation of contours

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

Let γ_0, γ_1 be closed contours in a region Ω , both defined on $[0, 1]$. We say that γ_0 is **homotopic** to γ_1 (**relative to Ω**), and write $\gamma_0 \approx \gamma_1$, if there exists a continuous map H from the unit square $Q = [0, 1] \times [0, 1]$ to Ω with the following properties:

$$H(t, 0) = \gamma_0(t), \quad 0 \leq t \leq 1;$$

$$H(t, 1) = \gamma_1(t), \quad 0 \leq t \leq 1;$$

$$H(0, s) = H(1, s), \quad 0 \leq s \leq 1.$$

Definition

The continuous mapping H is called a **homotopy** (or a **continuous deformation**). If γ_1 is a constant path, then we say that γ_0 is **homotopic to a point**.

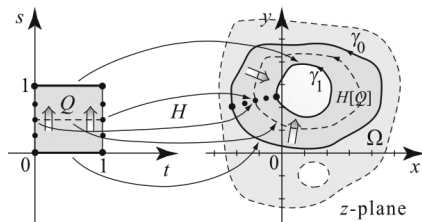


Figure 11: Continuous deformation of closed contours

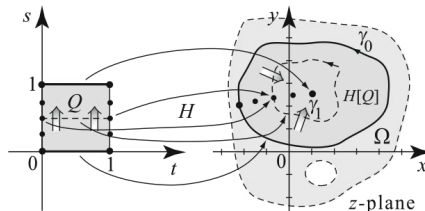


Figure 12: Continuous deformation to a point

Example 1.33

Let $\Omega = \mathbb{C}$. The upper semicircle $\gamma(t) = e^{it}, 0 \leq t \leq \pi$ in the figure shown below is homotopic to the directed line segment joining 1 to -1 .

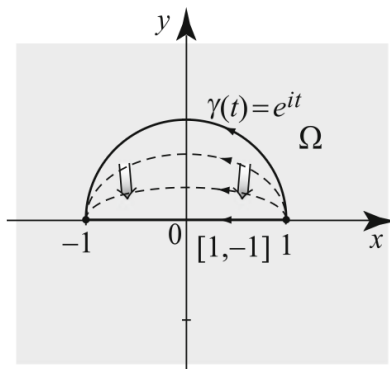


Figure 13: Semicircle γ

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Example 1.34

In the open unit disc $B_1(0)$, the polygonal path $[z_1, z_2, z_3]$ is homotopic to the line segment connecting z_1 to z_3 .

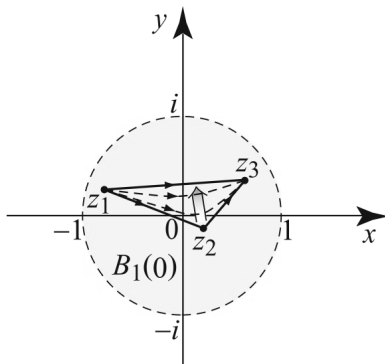


Figure 14: Polygonal path $[z_1, z_2, z_3]$

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Example 1.35

In the open unit disc $B_1(0)$, the arc γ_0 in the figure shown below is homotopic to the polygonal path $[z_1, z_2, z_3, z_1]$.

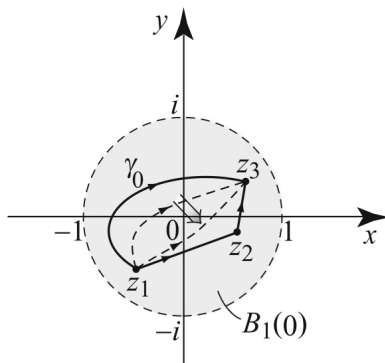


Figure 15: Arc γ_0

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Theorem 1.36 (Homotopic Version of Cauchy's Theorem)

Let Ω be a region, let $f \in \mathcal{H}(\Omega)$, and let γ, σ be contours in Ω , and Γ, Σ be closed contours in Ω .

i) If $\gamma \approx \sigma$, then

$$\int_{\gamma} f = \int_{\sigma} f.$$

ii) If $\Gamma \approx \Sigma$, then

$$\oint_{\Gamma} f = \oint_{\Sigma} f.$$

iii) If Γ is homotopic to a point in Ω , then

$$\oint_{\Gamma} f = 0.$$

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_{\gamma}$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Definition

A region $\Omega \subseteq \mathbb{C}$ is said to be **simply connected** if all contours joining α to β in Ω are homotopic relative to Ω . We will write $\Omega^\dagger$ to denote that $\Omega \subseteq \mathbb{C}$ is a simply connected region. If a region is not simply connected, then it is called **multiply connected**.

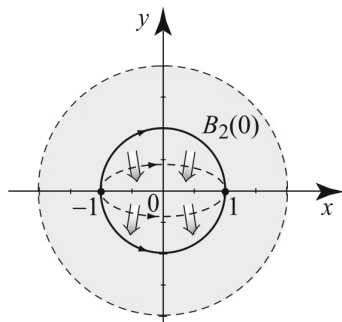


Figure 16: Simply connected region

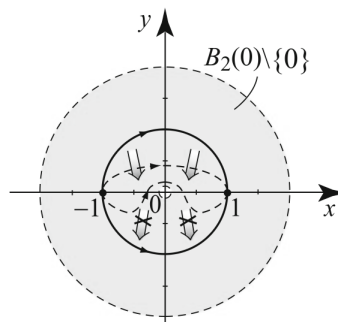


Figure 17: Multiply connected region

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Gauges

The Gauge Integral

Properties of $\int_a^b$ The Fundamental
Theorem (Calculus)The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$ The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

HomotopyCauchy's Integral
FormulaThe Fundamental
Theorem of Algebra

Taylor's Theorem

Theorem 1.37 (Cauchy-Goursat Theorem for Simply Connected Regions)

Let f be a holomorphic function on Ω° . If Γ is a closed contour in Ω° , then

$$\oint_{\Gamma} f = 0.$$

Corollary 1.38

Let f be holomorphic on Ω° . Then the following are valid:

- i) f has a primitive on Ω ;
- ii) If γ and μ are two contours in Ω with the same endpoints, then

$$\int_\gamma f = \int_\mu f.$$

Notation

Whenever the orientation of a closed contour is relevant, we will write

$$\oint_{\Gamma}^+ f \quad \text{and} \quad \oint_{\Gamma}^- f$$

to denote that Γ has a positive and negative orientation, respectively. When no orientation is specified, we will usually assume the positive orientation.

Theorem 1.39 (Cauchy-Goursat Theorem for Multiply Connected Domains)

Let $\{\Omega_k^{\circ}\}_{k=1}^n$ be pairwise disjoint, let their union be contained in Ω_0° , and let

$$\Omega = \Omega_0^{\circ} \setminus \left(\bigcup_{k=1}^n \overline{\Omega_k^{\circ}} \right).$$

For $j = 1, \dots, n$, let C_j be disjoint simple closed contours with positive [respectively, negative] orientation whose interior regions contain $\overline{\Omega_j^{\circ}}$ and assume that the C_j 's are pairwise disjoint. Let C be a simple closed contour with positive [respectively, negative] orientation in Ω whose interior region contains all C_j 's. Suppose that $f \in \mathcal{H}(\Omega)$. Then we have

$$\oint_C f = \sum_{i=1}^n \oint_{C_i} f \quad \left[\text{respectively, } \oint_C f = \sum_{i=1}^n \oint_{C_i} f \right].$$

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_{\gamma}$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

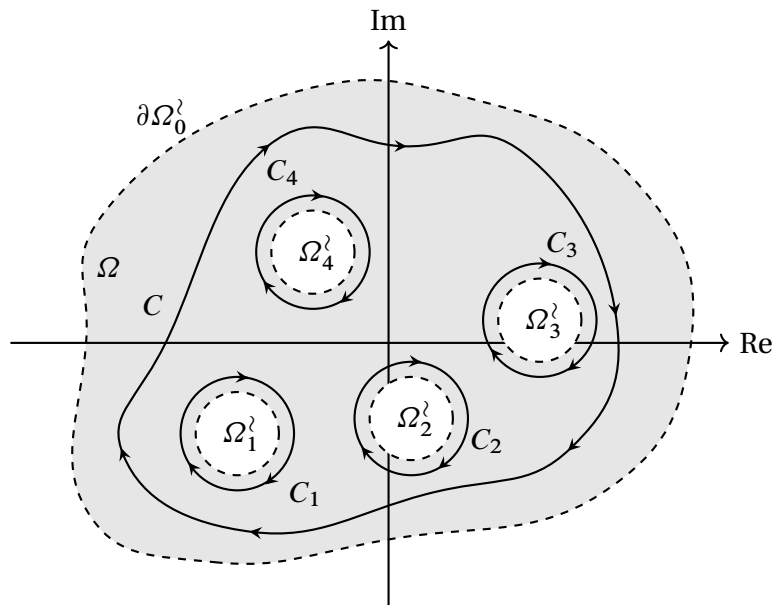


Figure 18: Geometric visualisation of the hypotheses in the Cauchy-Goursat theorem for multiply connected domains in the case $n = 4$

Gauges

The Gauge Integral

Properties of $\int_a^b$ The Fundamental
Theorem (Calculus)The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$ The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

HomotopyCauchy's Integral
FormulaThe Fundamental
Theorem of Algebra

Taylor's Theorem

Example 1.40

Let Γ be a positively oriented, simple, closed contour, and z_0 be a point not on $\Gamma^\dagger$. Then find a closed form expression for the integral

$$\oint_{\Gamma} dz (z - z_0)^n$$

for each integer n .

Example 1.41

Let C be a positively oriented, simple, closed contour containing i in its interior and $-i$ in its exterior. Evaluate

$$\oint_C dz \frac{1}{z^2 + 1}.$$

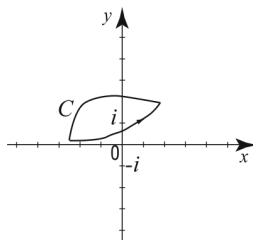


Figure 19: The contour C

Example 1.42

Evaluate the integral

$$\oint_C dz \frac{z}{(z-i)(z-1)},$$

where C is the figure-eight path shown in Figure 20

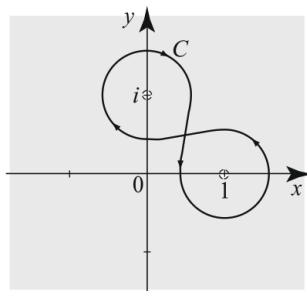


Figure 20: The (not simple) curve C

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Gauges

The Gauge Integral

Properties of $\int_a^b$ The Fundamental
Theorem (Calculus)The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$ The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

**Cauchy's Integral
Formula**The Fundamental
Theorem of Algebra

Taylor's Theorem

Theorem 1.43 (Cauchy Integral Formula)

Let $f \in \mathcal{H}(\Omega)$ with Ω a region that contains a simple closed contour Γ with positive orientation and its interior. If z_0 is a point in the interior of Γ , then

$$f(z_0) = \frac{1}{2\pi i} \oint_\Gamma dz \frac{f(z)}{z - z_0}.$$

Example 1.44

Compute

i) $\oint_{B_2(0)} dz \frac{e^z}{z+1}$

ii) $\oint_{B_2(1)} dz \frac{z^2 + 3z - 1}{(z+3)(z-2)}$

iii) $\oint_{B_2(0)} dz \frac{e^{\pi z}}{z^2 + 1}$

Lemma 1.45

Suppose that f is holomorphic on an open subset containing $\overline{B_R(z_0)}$ and satisfies $|f(z)| \leq M$ for all $z \in \overline{B_R(z_0)}$. Then for $0 < |z - z_0| < R/2$ the following are valid:

- i) $f'(z_0) = \frac{1}{2\pi i} \oint_{B_R(z_0)} d\zeta \frac{f(\zeta)}{(\zeta - z_0)^2};$
- ii) $\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| \leq 2M \frac{|z - z_0|}{R^2}.$

Theorem 1.46 (Leibniz Rule)

Let γ be a contour and let Ω be an open subset of $\mathbb{C}$. Let $\phi(z, \zeta)$ be a function defined for $z \in \Omega$ and $\zeta \in \gamma^\dagger$. Suppose that $\phi(z, \zeta) : \Omega \times \gamma^\dagger \rightarrow \mathbb{C}$ is continuous in ζ and holomorphic in z with a continuous derivative $\partial_z \phi(z, \zeta)$ in ζ . Then the function

$$g(z) = \int_\gamma d\zeta \phi(z, \zeta)$$

is in $\mathcal{H}(\Omega)$ and its derivative is

$$\partial_z g(z) = \int_\gamma d\zeta \partial_z \phi(z, \zeta).$$

Theorem 1.47 (Generalised Cauchy Integral Formula)

Let f be holomorphic on a region Ω containing a positively oriented simple closed contour Γ and its interior. Then f is infinitely differentiable at each $z \in \text{int } \Gamma$ and

$$\frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \oint_{\Gamma} dz \frac{f(z)}{(z - z_0)^{n+1}}.$$

Gauges

The Gauge Integral

Properties of $\int_a^b$ The Fundamental
Theorem (Calculus)The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$ The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

**Cauchy's Integral
Formula**The Fundamental
Theorem of Algebra

Taylor's Theorem

Example 1.48*Compute the following integral:*

$$\frac{1}{2\pi i} \oint_{B_2(0)} dz \frac{z^{10}}{(z-1)^{11}}.$$

Example 1.49

Let γ be the ellipse in Figure 21. Compute the following integral:

$$\int_{\gamma} dz \frac{e^{iz}}{(z - \pi)^3}.$$

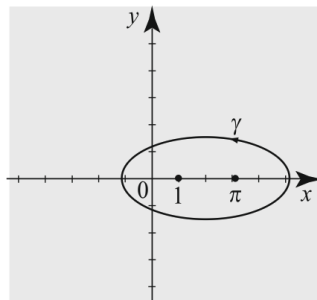


Figure 21: Ellipse with foci at 1, π

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_{\gamma}$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Remark

We can now complete the proof of the Cauchy-Riemann theorem, the following theorem is the remaining proof.

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental
Theorem (Calculus)

The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

**Cauchy's Integral
Formula**

The Fundamental
Theorem of Algebra

Taylor's Theorem

Theorem 1.50

If f is holomorphic in a domain Ω , then all partial derivatives of u and v exist and are continuous in Ω .

Theorem 1.51 (Mean Value Theorem)

Let $f \in \mathcal{H}(\Omega)$. Then for any disc $B_r(a)$ which together with its closure belongs to the region Ω , the value of f at the centre of the disc is equal to the mean value of this function taken around the disc boundary, i.e.

$$f(a) = \frac{1}{2\pi} \int_0^{2\pi} dt \, f(a + re^{it}).$$

Theorem 1.52 (Morera's Theorem)

Let f be continuous in a region Ω . Suppose that for all closed discs $\overline{B}$ contained in Ω and for all closed contours Γ in $\overline{B}$ we have

$$\oint_{\Gamma} f = 0.$$

Then $f \in \mathcal{H}(\Omega)$.

Theorem 1.53 (Cauchy's Inequality)

Let Γ be the positively oriented circle with centre z_0 and radius r and suppose f that is holomorphic inside and on Γ . If $|f(\Gamma)| \leq M$ for some positive number M , then

$$|f^{(n)}(z_0)| \leq \frac{Mn!}{r^n}, \quad n \in \mathbb{N}_0.$$

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental
Theorem (Calculus)

The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral
Formula

**The Fundamental
Theorem of Algebra**

Taylor's Theorem

Theorem 1.54 (Liouville's Theorem)

If f is a bounded and entire function, then it must be a constant.

Gauges

The Gauge Integral

Properties of $\int_a^b$ The Fundamental
Theorem (Calculus)The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$ The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral
Formula**The Fundamental
Theorem of Algebra**

Taylor's Theorem

Definition

A body $\mathbb{K}$ is said to be **algebraically closed** if every nonconstant polynomial with coefficients in $\mathbb{K}$ has a root in $\mathbb{K}$.

Gauges

The Gauge Integral

Properties of $\int_a^b$ The Fundamental
Theorem (Calculus)The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$ The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral
Formula**The Fundamental
Theorem of Algebra**

Taylor's Theorem

Remark

The real body $\mathbb{R}$ is not algebraically closed, for example, the polynomial

$$p(x) = x^2 + 1$$

has real coefficients but has no root in $\mathbb{R}$.

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental
Theorem (Calculus)

The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral
Formula

**The Fundamental
Theorem of Algebra**

Taylor's Theorem

Theorem 1.55 (The Fundamental Theorem of Algebra)

The set $\mathbb{C}$ of complex numbers is an algebraically closed body.

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental
Theorem (Calculus)

The Lebesgue
Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental
Theorem of Contour
Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral
Formula

**The Fundamental
Theorem of Algebra**

Taylor's Theorem

Corollary 1.56

Every polynomial of degree $n \geq 1$ has exactly n roots (up to multiplicities).

Theorem 1.57 (Maximum Modulus Principle)

Suppose that f is holomorphic on a region Ω . If $|f|$ attains a maximum on Ω , then f is constant in Ω .

Corollary 1.58

Suppose that Ω is a bounded region and that f is holomorphic on Ω and continuous on its closure $\overline{\Omega}$. Then

- i) $|f|$ attains its maximum M on $\partial\Omega$, and
- ii) either f is constant or $|f(z)| < M$ for all $z \in \Omega$.

Theorem 1.59 (Taylor's Theorem)

Let Ω be a domain, let $f \in \mathcal{H}(\Omega)$ and let z_0 be a point in Ω . Then $f \in \mathcal{A}(\Omega)$ with radius of convergence $R = \inf\{|z - z_0| : z \in \partial\Omega\}$, namely

$$f(z) = \sum_{n \in \mathbb{N}_0} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n, \quad \text{for } |z - z_0| < R.$$

This series representation is called the **Taylor expansion of f around z_0** .

Gauges

The Gauge Integral

Properties of $\int_a^b$

The Fundamental Theorem (Calculus)

The Lebesgue Integral

Parametrisations

Contour Integration

Properties of $\int_\gamma$

The Fundamental Theorem of Contour Integration

Cauchy's Theorem

Homotopy

Cauchy's Integral Formula

The Fundamental Theorem of Algebra

Taylor's Theorem

Corollary 1.60

If Ω is a domain in $\mathbb{C}$, then $\mathcal{H}(\Omega) = \mathcal{A}(\Omega)$.